Fiscal Year:	FY 2004	**Task Last Updated:**	FY 06/14/2004
PI Name:	Benard, Andre Ph.D.		
Project Title:	Investigation of Solidification Processes with Convection Using Meshless Methods and Quantitative Experimental Verification		
Division Name:	Physical Sciences		
Program/Discipline:	MATERIALS SCIENCE		
Program/Discipline--Element/Subdiscipline:	MATERIALS SCIENCE--Basics of solidification		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	East Lansing	**State:**	MI
Zip Code:	48824	**Congressional District:**	8
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	98-HEDS-05
Start Date:	05/16/2000	**End Date:**	11/30/2003
No. of Post Docs:	0	**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA MSFC
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Diaz, Alejandro (Michigan State University) Koochesfahani, Manoochehr (Michigan State University) McGrath, John (University of Arizona)		
Grant/Contract No.:	NAG8-1690		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	The experimental work consists first of making quantitative measurements of the velocity field, and subsequently temperature field, during the solidification of an optically transparent analog of a binary metallic alloy. A novel Molecular Tagging Velocimetry (MTV) technique is used to perform the velocity field measurements. Attempts are also in progress to combine MTV with Laser-Induced Fluorescence (LIF) in order to perform temperature measurements. Such quantitative measurement of the velocity field during the solidification process will enhance our understanding of the diverse transport phenomena occurring during solidification and permit the refinement of the phenomenological models used in averaged transport equations. The experimental data will be compared with the results of solving the averaged governing equations. The objectives of the numerical research are to use a fictitious domain approach to solve averaged governing equations describing the overall solidification process. The use of this method will reduce the time required to generate models by eliminating the need for cumbersome mesh generation. The method is also highly efficient for large problems. The model will be tested against the experimental data, as they become available, and improved as needed. At the end of this program, a new computational tool that can perform efficient computations for solidification processes will be implemented and validated. Novel experimental and numerical methods are applied in this work to study solidification processes in order to produce an improved fundamental understanding of solidification processes, including the role played by buoyancy-driven convection phenomena in determining the microstructure. The use of MTV/LIF to study transport phenomena in transparent systems (solidification processes in this project) will provide novel experimental results spanning various cavity sizes and chilled configurations. This will demonstrate the capabilities of the MTV/LIF method, provide verification of the modeling approaches proposed, and provide benchmark velocity and temperature field data valuable to other analytical and numerical studies, as well as enhance our understanding of thermo-solutal convection processes. It may also lead to unprecedented micro-scale field measurements around the mushy zone, which would lead to the refinement of the mushy zone models. Finally, this research might lead to the development of a MTV/LIF method that is adapted for use in space, as a module on board the Materials Science Research Facility.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	Laser-Induced-Fluorescence temperature measurements were performed in the test bottom chilled cavity. A two-color LIF approach based on using one dye and two-dyes were developed. Experiments performed with two dyes were shown to improve significantly the accuracy of the temperature measurements. Various measurements were performed above chimneys and the entire mushy region of the test cavity.
Bibliography Type:	Description: (Last Updated: 02/25/2019)